

CHEM 408 - Introduction to Green Chemistry - Fall 2024

Asynchronous Online Lecture

Instructor – Dr. Sarah Prescott (she/her)

Preferred Contact - Canvas inbox or Teams

Teams Course Channel (optional)

[Dr. Prescott's Classes | Green Goggles Fall 2024 | Microsoft Teams](#)

Office hours – Thursdays 10-11 am ET in person or Virtual on Teams to ask any questions or get additional assistance. Other times by request and based on availability.

Course Materials

We will use the annotation tool Perusall in this course, which has our course text embedded within it and is also where you will complete your problem-solving assignments. More details are provided in the course orientation.

Important!

This course is designed to create a deeper understanding of the matter and energy that make up the chemical aspects of our world through the lens (goggles) of green chemistry. My hope is that you leave this class feeling accomplished, challenged, and excited about learning more. However, your safety and well-being are more important than this class. If you are facing any medical, family, housing, or food security challenges know that you can reach out and have support available (see links later in the syllabus). If you are comfortable, I hope that you will reach out to me as well and I will help you with anything I can.

My Promises

- I'm on your team - I want you to be successful and am willing to help you figure out what you need and I care about your goals.
- I will give you the tools and space to learn. I work very hard to figure out and understand the ways people learn best, embed those ways and

structures in all my courses, and give you time to learn. I don't ask you to do things that do not generate a better understanding and connection to your learning. It also means that I challenge you to LEARN and not just perform tasks for a grade.

- I will actively listen and encourage you to give me feedback and communicate openly. I make changes to my courses every time I teach them in response to my own reflection along with student feedback and ideas.

Course Catalog Description

CHEM 408 (M1) - Green Goggles - Introduction to Green Chemistry

Term: Fall 2024 - Full Term (08/26/2024 - 12/09/2024)

Credits: 4.0

CRN: 14134

In this course, we investigate the principles and practice of Green Chemistry as they connect to real world examples. Green Chemistry is the field of science that uses a principle-based approach to design chemical reactions and processes to make them more sustainable. In exploring green chemistry, many of the fundamental concepts of a general chemistry course are investigated. Some topics include use of renewable feedstocks, atom economy, catalysis, waste prevention, and design for degradation.

Equivalent(s): CHEM 444G

Discovery Course

The faculty aspire for students to contextualize their major fields of study within the broader contexts of the world's challenges and opportunities. The Discovery Program, UNH's undergraduate core curriculum, supports that outcome by ensuring that as part of their bachelor's degrees, all students understand and explore a breadth of disciplinary content areas. Please note that this course meets the Physical Science Discovery Category requirement.

Student Learning Outcomes

Upon successful completion of this course, students will have the opportunity to:
(*italicized items also meet outcomes for the Physical Science Discovery Category*).

- *Discuss and analyze the everyday chemistry occurring around them*
 - *Demonstrate understanding of fundamental chemical principles using the principles and practice of Green Chemistry*
 - Understand the integration of the principles and practice of Green Chemistry in a variety of chemical applications
 - Disseminate discovery of the world through “green goggles” to the larger community
 - *Continue to become more scientifically literate and use computational thinking to understand chemistry concepts and how they apply to a real-world context.*
 - Analyze scientific research and report on specific findings relevant to a larger non-scientific community, using digital and information literacy skills.
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Grading

Scale

Final grades in this course will be based on the following scale: 94-100=A; 90-93.9=A-; 87-89.9=B+; 83-86.9=B; 80-82.9=B-; 76-79=C+; 73-75=C; 70-72.9=C-; 66-69.9=D+; 63-65.9=D; 60-62.9=D-; Below 59.9=F.

Distribution

You can access your grades at any time via Canvas/Perusall. Below is a percentage distribution of all components toward your final course grade. An outline of each component is included in this syllabus. See course modules for more detailed information on each component. Please check your grades regularly and reach out to me with any questions. The distribution of main course components is highlighted below. See the course site for further details.

Perusall Assignments	30%
Discussion Boards	30%
Quizzes	30%

Course Structure/Format

Asynchronous Online Format

The course is run in an asynchronous online format, meaning we do not meet together online for the entirety of the course. Instead, you will be working on self-guided interactive modules that are completed in order every week. It is the student's responsibility to read all class materials and complete assignments on time. As there is no formal class session, you must keep up with all assigned tasks. Course content has been specifically designed and chosen to aid in your learning in the course. Students who are most successful in this type of course are those who are motivated to learn and willing to learn independently.

- It is the student's responsibility to have access to a desktop/laptop computer and reliable internet connection for this course. A tablet device or phone is insufficient to complete the work.
 - My role as your instructor throughout the semester will be to act as a guide and mentor, answering questions you have to support and encourage your learning.
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Course Assignments/Assessments

General Notes

- You may print out assignments and course materials for your use.
- An outline of assignments is below, but more details on all course assignments are contained on the course site.
- All assignments must be solely your own work. The use of AI-assisted writing (such as ChatGPT) is strictly prohibited on any assignment in this course unless otherwise instructed.
- Late assignments do not receive credit unless you have made arrangements with me ahead of time. That said -- please reach out if you are falling behind -- I understand that life happens and will do my best to work with you.

Perusall - Assigned Reading and Problem posing and solving - 30%

- You will use the annotation tool Perusall to complete assigned text reading and problem-solving activities in weekly assignments
- Enhances your learning experience, and allows you to engage and learn with others in a supportive learning environment
- Overall average score used for course grade - see Perusall site throughout the course for your individual assignment and running average scores
 - You must pass this course component overall with a 60% or higher score to pass the course.

Discussion Boards -Community Engagement and Learning Application/Reflection - 30%

- Assignments will allow you to apply what you are learning in the course and share your discoveries with your peers
- Weekly entries will also serve to help you reflect upon your learning in the course
- Used to evaluate formative understanding of concepts in the course
- Evaluated for grading periodically
 - Late submissions will reduce the grade.

Quizzes – 30%

- Evaluate your understanding of key concepts presented in the course and give feedback on your learning.
- Two attempts on the first quiz only (highest score recorded), all others a single attempt.
- Lowest two scores will automatically be dropped from computing overall course grade.

Final Assessment-10%

- A take-home final assessment with student choice to give the opportunity to research and synthesize your learning.
- Designed to evaluate your summary understanding of fundamental course concepts.
- Multiple options to choose from to demonstrate your synthesis of course content and ability to apply the content
 - All options require accurate content, proper formatting, clear communication, fair use, and connection to green and general chemistry knowledge

- Additional details will be provided and it will be open for submission following mid semester and will be due Monday, December 9th. You are encouraged to submit this portion of the exam as early as you like.
 - You must pass this course component overall with a 60% or higher score to pass the course.
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Course Policies

I will highlight some important information specifically about this course in this next section, followed by our course overall schedule. Following the schedule is more information that is general to most UNH courses. I hope you find this organization to be more convenient and practical for you to understand what this course entails.

Class Cancellation

As this is an asynchronous course, if UNH Manchester cancels classes, it does not affect this course. Due dates will remain in effect regardless of the face-to-face courses' status.

Academic Conduct

You are encouraged to share with classmates and others the discoveries you are making and collaborative learning is built into the format of this course. However, when you submit answers to any assignments, make sure all answers are in your own words and express your own thoughts. The use of AI-assisted writing (such as ChatGPT) is strictly prohibited on any assignment in this course unless otherwise instructed.

Submitting another's work as your own is plagiarism and may result in course failure and possible further administrative action. Please ask me if you have any questions.

Courtesy and respect towards your classmates and the instructor will always be required. All online discussions must be done in a courteous and respectful manner.

Course Technology and Requirements

- You will need to have a desktop/laptop and reliable internet access for this course.
- You will need a scientific calculator in this course, that can perform calculations using exponents and scientific notation.

- Canvas and Perusall will be used extensively in this course. Zoom (or Teams) may also be used.
 - Canvas will be used to post weekly course announcements and contain course information. For any formal course communications (grades, accessibility, etc.) please use the canvas inbox to message me.
 - Perusall is where you will access your text and weekly reading and problem-solving assignments.
 - An orientation to the onboarding and use of Perusall are provided in the course orientation module.
 - Teams is an optional way for you to send messages and questions to me.
- You will need to be able to use general online technologies including: Google Docs and Sheets, varying file types (.doc, .docx, .pdf), website searching and navigation, and capturing screenshots. Supplemental resources and tutorials are provided on the course site to assist you. If you need additional assistance, drop in to office hours or message me and I am happy to assist.
- If you wait until just before the deadline to submit your work, then you assume the risks associated with technological problems or lack of familiarity with the platform. Please plan ahead and save your work in a backup form. I recommend using Google Drive or other cloud-based drives that autosave as you work. Google Docs are a great way to draft any posts or assignment information so that you can copy/paste into whatever format is needed.

CHEM 408 Fall 2024 Schedule

The schedule below is subject to change at instructor's discretion.

Week	Lecture Topic	Week	Lecture Topic
1	Intro to Green Chemistry	9	Bio- and Geo-spheres

2	Atoms and Elements	10	Organic and Bio-Chemistry
3	Compounds Part 1	11	Energy
4	Compounds Part 2	12	GC in your life!
5	Reactions Part 1	13	Industrial Ecology
6	Reactions Part 2	14	Thanksgiving
7	Water, solvents, solutions	15	Toxicology
8	Air and other gases		

Additional Policies, Resources, and Information

Temporary Academic Supports for Extended Absences

If you are dealing with an unexpected, extenuating circumstance that will keep you out of class or affect your performance for more than a day or two, reach out to Lisa Enright, Assistant Dean of Student Success, at lisa.enright@unh.edu to request a letter be sent to all your faculty.

Mental Health and Wellness

Your academic success and overall mental health are very important. If, during the course, you find you are experiencing emotional or mental health issues, please reach out. In partnership with The Mental Health Center of Greater Manchester, UNH

Manchester offers free mental health sessions for students. You may email unhm.wellness@unh.edu to make an appointment to meet with a counselor.

The National Suicide Prevention Lifeline provides 24/7, free and confidential support via phone or chat for people in distress, resources for you or your loved ones, and best practices for professionals. Call (800) 273-TALK (8255).

Here is a [list of other general resources](#).

Basic Needs

It can be challenging to do your best in class if you have trouble meeting basic needs like safe shelter, sleep, and nutrition. If you have difficulty affording groceries or accessing sufficient food to eat every day or lack a safe and stable place to live, I urge you to contact The Dean of Students and/or me if you are comfortable. You can also find a [list of resources](#). We are here to help.

NameCoach

We have a new tool in Canvas, NameCoach, that will help us all to pronounce each other's names properly. Please join me in making our learning environment as inclusive as possible by recording your name (instructions [here](#)) and taking advantage of the opportunity to respectfully learn how to pronounce each other's names.

Needs of Students of all Faiths

In the event that you need accommodation for a religious or cultural holiday/observance, please let me know as early as possible.

Academic Alerts

The University is invested in your academic success. If I am concerned about your academic behavior or performance, I may submit an academic alert. Academic alerts are not punitive. The goal is to provide you with support and resources to support your success. They act as an important check-in point and, if you receive an academic alert, you will receive an email to your UNH email address. It is strongly recommended that you meet with a professional advisor and connect with me to discuss the reason for the alert.

Netiquette Guidelines

Netiquette is the socially and professionally acceptable way to communicate on the internet. We are all expected to abide by these guidelines of "netiquette" when using

online communication tools with your classmates and instructor. Guidelines can be found on the [Student Resource Hub](#) and in the course orientation materials.

Academic Honesty

The Biology Program at UNH Manchester will strictly adhere to the University policy on academic honesty, as published in the UNH Student Rights, Rules, and Responsibilities Handbook (<http://unh.edu/vpsas/handbook/academic-honesty>). By turning in any piece of work in this course, you declare that you have read and understood the policy and that you did not engage in any form of academic dishonesty as defined in the Handbook.

Plagiarism can take many forms, such as: submitting someone else's work - in whole or in part - as your own; collaborating on answers for individual assignments or allowing your own work to be used by another student; copying information from a website or other text without proper documentation; buying a pre-written paper or lab report.

Cheating is mainly concerned with copying on exams or in the lab, bringing crib notes into an exam, or referring to notes or the textbook or any other source such as a programmable calculator, tablet, or cell phone during an exam.

Any instances of cheating or plagiarism will result in consequences that can range from a failing grade on the assignment for all students involved to dismissal from the University, as defined in the UNH Student Rights, Rules, and Responsibilities Handbook.

Any form of academic misconduct on any graded work in this course (including assignments, quizzes, or exams) will result in an automatic zero on the assignment. Additional consequences may result at the discretion of the instructor.

UNH Land, Water, and Life Acknowledgement

As we all journey on the trail of life, we wish to acknowledge the spiritual and physical connection the Pennacook, Abenaki, and Wabanaki Peoples have maintained to N'dakinna (homeland) and the aki (land), nebi (water), olakwika (flora), and awaasak (fauna) which the University of New Hampshire community is honored to steward today. We also acknowledge the hardships they continue to endure after the loss of unceded homelands and champion the university's responsibility to foster relationships and opportunities that strengthen the well-being of the Indigenous People who carry forward the traditions of their ancestors.

Credit Hour Compliance

This syllabus reflects the federal definition of a credit hour, which entails a minimum of 3 hours of engaged time per week per credit over the course. Examples of engaged time include class time, assignments, examinations, laboratories, participation in course-related experiences (attending a talk or performance, speakers and events, fieldwork, etc.), conferences, and office hours.

Class Recording

Class lectures and activities may be recorded in order to make this course more accessible, and recordings may be made available to students presently enrolled in the class to assist in reviewing material and preparing for assessments. Recordings are to be used for educational use/purposes only. Students may not record class activities (including screenshots showing instructors and students) without express permission from their instructor. If permission is granted, student-initiated recordings may only be used for educational purposes in this class and only during the period in which the student is enrolled in the class; authorized student-initiated recordings may not be posted or shared in any fashion outside of the class. The University and Zoom have FERPA-compliant agreements in place to protect the security and privacy of UNH Zoom accounts. Students may not share recordings outside of their course.

Confidentiality and Mandatory Reporting

The University of New Hampshire and its faculty are committed to assuring a safe and productive educational environment for all students and for the university as a whole. To this end, the university requires faculty members to report to the university's [Title IX Coordinator](#) (Bo Zaryckyj, Bo.Zaryckyj@unh.edu, 603-862-2930/1527 TTY) or to the UNH Manchester/CPS Title IX Deputy Intake Coordinator (Lisa Enright, lisa.enright@unh.edu 603-641-4336, Room 439) any incidents of sexual violence and harassment shared by students.

If you wish to speak to a confidential support service provider who does not have this reporting responsibility because their discussions with clients are subject to legal privilege, you can contact the [SHARPP Center for Interpersonal Violence Awareness, Prevention, and Advocacy](#) at (603) 862-7233/TTY (800) 735-2964. For more information about what happens when you report, how the university treats your information once a report is made to the Title IX Coordinator, your rights and reporting options at UNH (including anonymous reporting options) please visit [student reporting options](#). The [uSafeUS app](#) is also available for students to keep reporting options and resources easily accessible on their phones.

Help us improve our campus and community climate. If you have observed or experienced an incident of bias, discrimination or harassment, please report the incident

by contacting the Civil Rights & Equity Office at UNH.civilrights@unh.edu or TEL # (603) 862-2930 voice/ (603) 862-1527 TTY / 7-1-1 Relay NH, or [visit the CREO website](#). Anonymous reports may be submitted.

Confidential Support Resources for UNH Manchester:

- SHARPP Extended Services Coordinator for UNH Manchester- room 417. Available in person Mondays 9 am to 4-pm and by Zoom appointment year-round M-F.
- YWCA, NH – 603-668-2299(24hour), 72 Concord St. Manchester, NH
- The Mental Health Center of Greater Manchester: On campus mental health counselors are available by appointment during the academic year. Located in room 426.
- 24 Hour NH Sexual Violence Hotline: 1-800-277-5570
- 24 Hour NH Domestic Violence Hotline: 1-866-644-3574

Accessibility Services

According to the Americans with Disabilities Act (as amended, 2008), each student with a disability has the right to request services from UNH to accommodate his/her/their disability. If you are a student with a documented disability or believe you may have a disability that requires accommodations, please contact Student Accessibility Services (SAS) located on the Manchester campus in the Student Services Suite (Office 405A). Accommodation letters are created by SAS with the student. Please follow-up with me as soon as possible to ensure timely implementation of the identified accommodations in the letter. Faculty have an obligation to respond once they receive official notice of accommodations from SAS but are under no obligation to provide retroactive accommodations. For more information refer to www.unh.edu/sas or contact SAS at 603.862.2607, 711 (Relay NH) or sas.office@unh.edu.

Additional Resources

Library

The UNH Manchester librarians are available to assist you with the research process. Visit the library's website at <https://cps.unh.edu/library> to learn about library services and to search for reliable academic sources. You can contact the library at 603-641-4173 or at unhm.library@unh.edu.

The links below guide you to useful online library resources:

- Make a Research Appointment with a librarian:
<https://libraryguides.unh.edu/remotearchive/researchhelp>
- Use the Library Search Box to find information:
https://libraryguides.unh.edu/librarysearchbox_unhmanchester
- Reserve a Study Room: <https://cps.unh.edu/library/support-services>
- Discover resources for Citing Sources:
<https://libraryguides.unh.edu/unhmcitingsources>
- Learn strategies for Evaluating Sources:
<https://libraryguides.unh.edu/ENGL401UNHManchester/evaluatingresources>.

Center for Academic Enrichment (CAE)

The Center for Academic Enrichment (CAE) is a free resource for all students enrolled in UNH Manchester courses, as well as alumni with writing for graduate school applications and career seeking. *ALL UNH Manchester students may receive 1 hour of individual tutoring per week for each course in which they are enrolled* (dependent on tutor availability). CAE tutors support most courses and writing across the curriculum. In addition, tutors can work with you around time management, study skills, note-taking and academic organization strategies. Tutors work by appointment only, so please plan ahead for the next day or beyond. Appointments are available Monday through Friday, with occasional night and weekends. Visit the [CAE myCourses](#) site for more information about our services, and to make an appointment. *To make an appointment call the CAE at (603) 641-4113 or directly access our [TracCloud](#) scheduling software.*

Food Pantry

The campus food pantry has pre-packed bags of non-perishable food items for students in room 437. If you have any questions, contact Lisa Enright.

Should you have any questions regarding any of the above policies, it is your responsibility to contact me to discuss any concerns.